

Mechanical Ventilator Operating and Maintenance Manual

Sunrise Medical Center

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Policy Owner: Director of Respiratory Therapy, James Okonkwo

Distribution: All Sunrise Medical Center Staff

1. Purpose

This manual provides Respiratory Therapy and ICU nursing staff with standardized guidance for the setup, operation, alarm management, and preventive maintenance of the mechanical ventilators deployed throughout the Intensive Care Unit and Emergency Department at Sunrise Medical Center.

2. Pre-Use Checklist

Before connecting any ventilator to a patient, the assigned Respiratory Therapist must complete the pre-use self-test cycle and visually confirm circuit integrity, humidifier water level, and filter placement. A ventilator that fails self-test must be immediately tagged out of service and reported to Biomedical Engineering; a backup unit must be retrieved from the equipment pool before proceeding.

- 1 Run the built-in ventilator self-test and confirm pass status on the display.
- 2 Inspect the breathing circuit for cracks, disconnection points, or moisture pooling.
- 3 Verify the bacterial/viral filter is within its labeled expiration and change interval.
- 4 Confirm the humidifier chamber is filled to the appropriate level and heater wire is connected.
- 5 Set initial alarm parameters per physician order before connecting the patient.

3. Alarm Management

High-priority alarms (apnea, disconnection, high pressure) must never be silenced for longer than 2 minutes without direct bedside evaluation of the patient and circuit. All alarm limit changes must be documented in the ventilator flow sheet with the ordering provider noted.

- High-pressure alarm: check for kinked tubing, biting, secretions, or worsening lung compliance.
- Low-pressure/disconnect alarm: check all circuit connections immediately, including the humidifier chamber and inline suction port.
- Apnea alarm: verify patient is breathing or receiving mandatory breaths; escalate immediately if unresponsive.

4. Circuit Change and Infection Control

Ventilator circuits are changed only when visibly soiled or malfunctioning, not on a fixed calendar schedule, per current CDC guidance on ventilator-associated pneumonia prevention. Heat-moisture exchanger filters are changed every 24 hours or sooner if saturated. All circuit changes are documented in the electronic health record with date, time, and RT initials.

5. Weaning and Extubation Readiness

Daily spontaneous awakening trials and spontaneous breathing trials are coordinated between nursing and Respiratory Therapy per the ICU Liberation Protocol (SMC-CC-011). Ventilator settings during a spontaneous breathing trial must be documented separately from baseline settings to avoid confusion during shift handoff.

6. Preventive Maintenance Schedule

Task	Frequency	Performed By
Full self-test and calibration check	Every use (pre-connection)	Respiratory Therapist
Deep clean and disinfection between patients	Every patient turnover	Respiratory Therapy / EVS
Preventive maintenance service	Every 12 months or per manufacturer hours	Biomedical Engineering
Battery backup function test	Monthly	Biomedical Engineering

7. Equipment Malfunction Reporting

Any suspected malfunction, unexpected alarm behavior, or patient safety event involving a ventilator must be reported immediately to Biomedical Engineering and documented via the Hospital Incident Reporting System within the same shift. The unit is quarantined and tagged "Do Not Use" pending investigation; it is never returned to circulation without Biomedical Engineering sign-off.

8. Related Documents

SMC-CC-011 ICU Liberation and Sedation Protocol; SMC-IC-008 Ventilator-Associated Pneumonia Prevention Bundle; SMC-BME-004 Medical Equipment Incident Reporting Procedure.

Appendix A: Frequently Asked Questions

Q: What should be done if a backup ventilator is also unavailable during a self-test failure? A: Escalate immediately to the Respiratory Therapy supervisor and Biomedical Engineering; manual bag-valve-mask ventilation is used to bridge the patient until a functioning unit is obtained, this should take only a few minutes given equipment pool stock levels.

Q: How quickly must Biomedical Engineering respond to a ventilator malfunction report? A: Within 15 minutes for an in-use ventilator with a patient still connected, and within one shift for a malfunction identified during pre-use testing on a standby unit.

Q: Can circuit changes be delayed if the unit is extremely busy? A: No — visibly soiled or malfunctioning circuits are changed regardless of unit workload, as this is a direct infection control and patient safety requirement, not a discretionary task.

Appendix B: Common Scenarios

Scenario — Recurrent high-pressure alarms overnight: The therapist should systematically rule out secretions, biting, and circuit kinking before adjusting alarm limits; alarm limits should never be widened simply to stop the alarm without identifying the underlying cause.

Scenario — Ventilator transported with patient to imaging: A full pre-transport checklist is completed including battery charge level, portable oxygen supply calculation, and backup bag-valve-mask availability, per the Intra-Hospital Transport Policy.

Revision History

Version	Date	Description
1.0	Jan 10, 2021	Initial release
1.2	Mar 18, 2025	Updated circuit change policy to align with CDC guidance